

TidyTuesday Week 41

Pranay Gundam

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling installment of *Correlations Nobody Asked For*, where we take two economic time series that have never met, sit them down at the same table, and force them to explain their relationship to a judgmental linear regression. This week's randomly selected odd couple: the Producer Price Index for Residential Natural Gas in the Middle Atlantic, and – deep breath – the Percent of Value of Loans Made Under Participation or Syndication for All Commercial and Industry Loans, Other Risk (Acceptable), U.S. Branches and Agencies of Foreign Banks, which is also, I should mention, DISCONTINUED. Nothing says rigorous science like one of your variables having been quietly retired in 2017.

The raw regression came out swinging. An R-squared of 0.44, a p-value of 0.001, a confidently negative coefficient – by the breathless standards of this project, these are practically wedding bells. "Natural gas prices go up, foreign bank syndicated loan risk percentages go down!" I can already hear the cable news chyron. Someone alert the Federal Reserve. Somebody write a thesis. The two series were apparently drifting in roughly opposite directions over their shared years, and the regression, bless its credulous little heart, decided that meant something.

And then we detrended it, and the whole romance evaporated like morning dew. Once we stripped out each series' own long-run march through time, the R-squared collapsed to a majestic 0.019, the p-value ballooned to 0.56, and the coefficient's confidence interval cheerfully straddles zero like it's got nothing to hide – because it doesn't. In other words, the "relationship" was pure shared drift: two trends passing in the night, mistaken for love. This is the textbook trend-mirage this project exists to gently mock, and it delivered one right on schedule.

As always, the mandatory reminder: these are randomly paired series with zero theoretical reason to be related, and nothing here constitutes a causal claim, a statistical finding, or financial advice you should act on in any capacity whatsoever. Spurious correlation isn't the bug here – it's the entire genre. This week's natural gas and discontinued foreign-bank loans were never a couple; they just happened to be walking the same direction for a while. See you next week for more statistical theater.

2 Date: 2026-10-06

Series ID: PCU2212102212101122

This series is titled Producer Price Index by Industry: Natural Gas Distribution: Residential Natural Gas for Middle Atlantic Census Division and has a frequency of Monthly. The units are Index Jun 2012=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-06-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: EFAOXFBNQ

This series is titled Percent of Value of Loans Made Under Participation or Syndication for All Commercial and Industry Loans, Other Risk (Acceptable), U.S. Branches and Agencies of Foreign Banks (DISCONTINUED) and has a frequency of Quarterly, 2nd Month's 1st Full Week. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2012-07-01 and the observation end date is 2017-04-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_EFAOXFBNQ	R-squared:	0.444
Model:	OLS	Adj. R-squared:	0.413
Method:	Least Squares	F-statistic:	14.37
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.00134
Time:	17:29:15	Log-Likelihood:	-69.835
No. Observations:	20	AIC:	143.7
Df Residuals:	18	BIC:	145.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	104.3514	21.206	4.921	0.000	59.798	148.904
value_fred_PCU2212102212101122	-0.8266	0.218	-3.790	0.001	-1.285	-0.368

Omnibus:	2.921	Durbin-Watson:	1.600
Prob(Omnibus):	0.232	Jarque-Bera (JB):	1.265
Skew:	0.146	Prob(JB):	0.531
Kurtosis:	1.803	Cond. No.	1.10e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

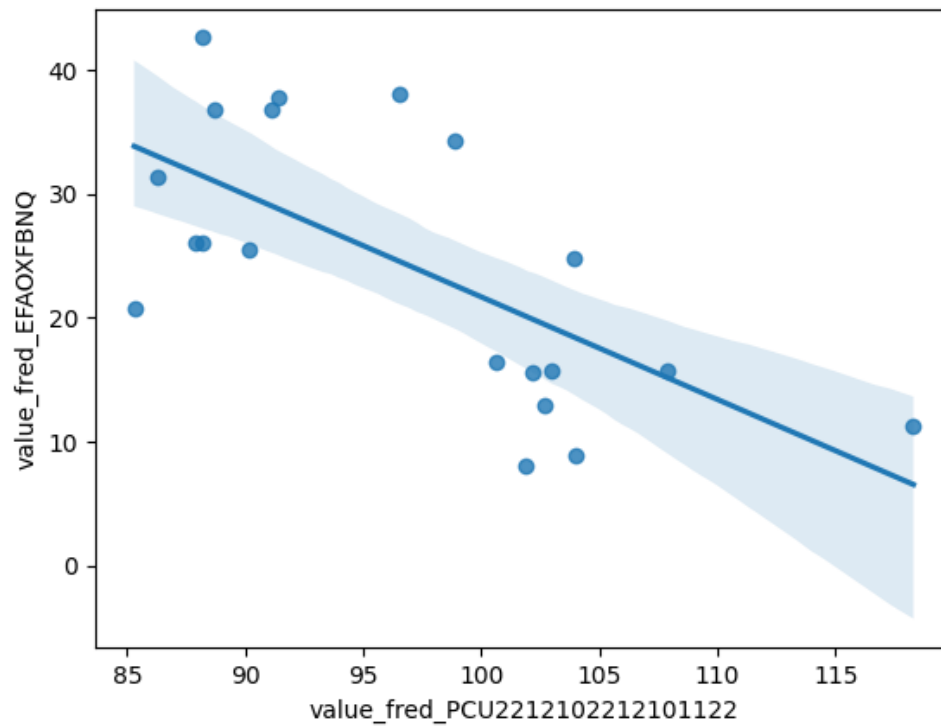


Figure 1: Raw Regression Plot for 2026-10-06

[2] The condition number is large, $1.1e+03$. This might indicate that there are strong multicollinearity or other numerical problems.

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_EFAOXFBNQ_detrended	R-squared:	0.019
Model:	OLS	Adj. R-squared:	-0.035
Method:	Least Squares	F-statistic:	0.3538
Date:	Tue, 06 Oct 2026	Prob (F-statistic):	0.559
Time:	17:29:15	Log-Likelihood:	-69.114
No. Observations:	20	AIC:	142.2
Df Residuals:	18	BIC:	144.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-4.379e-12	1.807	-2.42e-12	1.000	-3.796	3.796
value_fred_PCU2212102212101122_detrended	-0.2983	0.502	-0.595	0.559	-1.352	0.755

Omnibus:	0.524	Durbin-Watson:	1.639
Prob(Omnibus):	0.770	Jarque-Bera (JB):	0.580
Skew:	0.060	Prob(JB):	0.748
Kurtosis:	2.175	Cond. No.	3.60

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

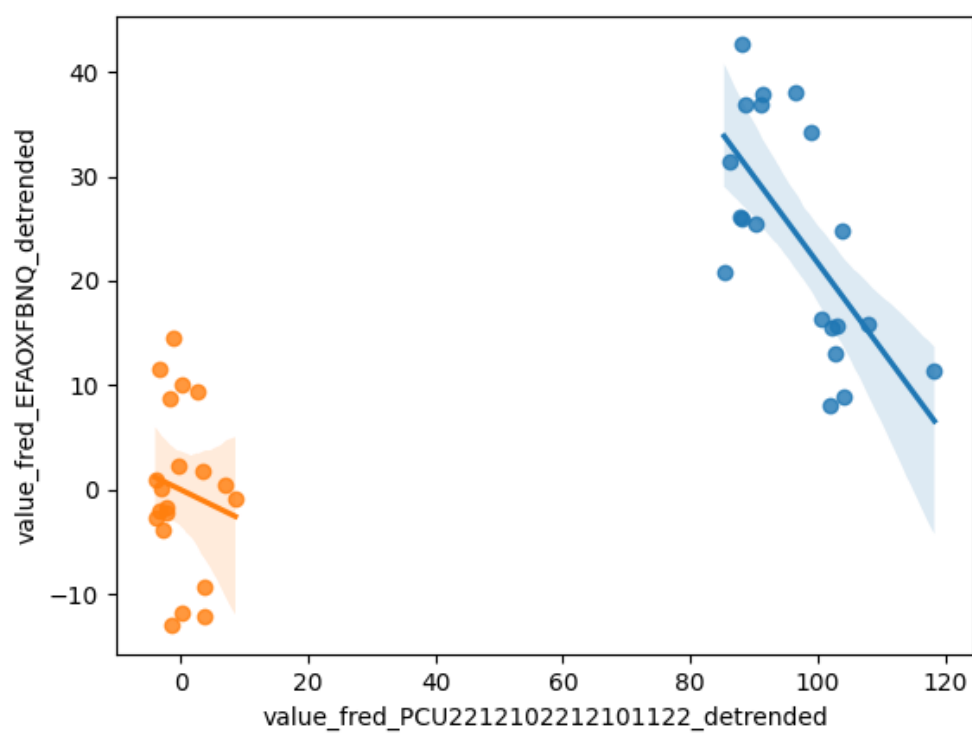


Figure 2: Detrended Regression Plot for 2026-10-06